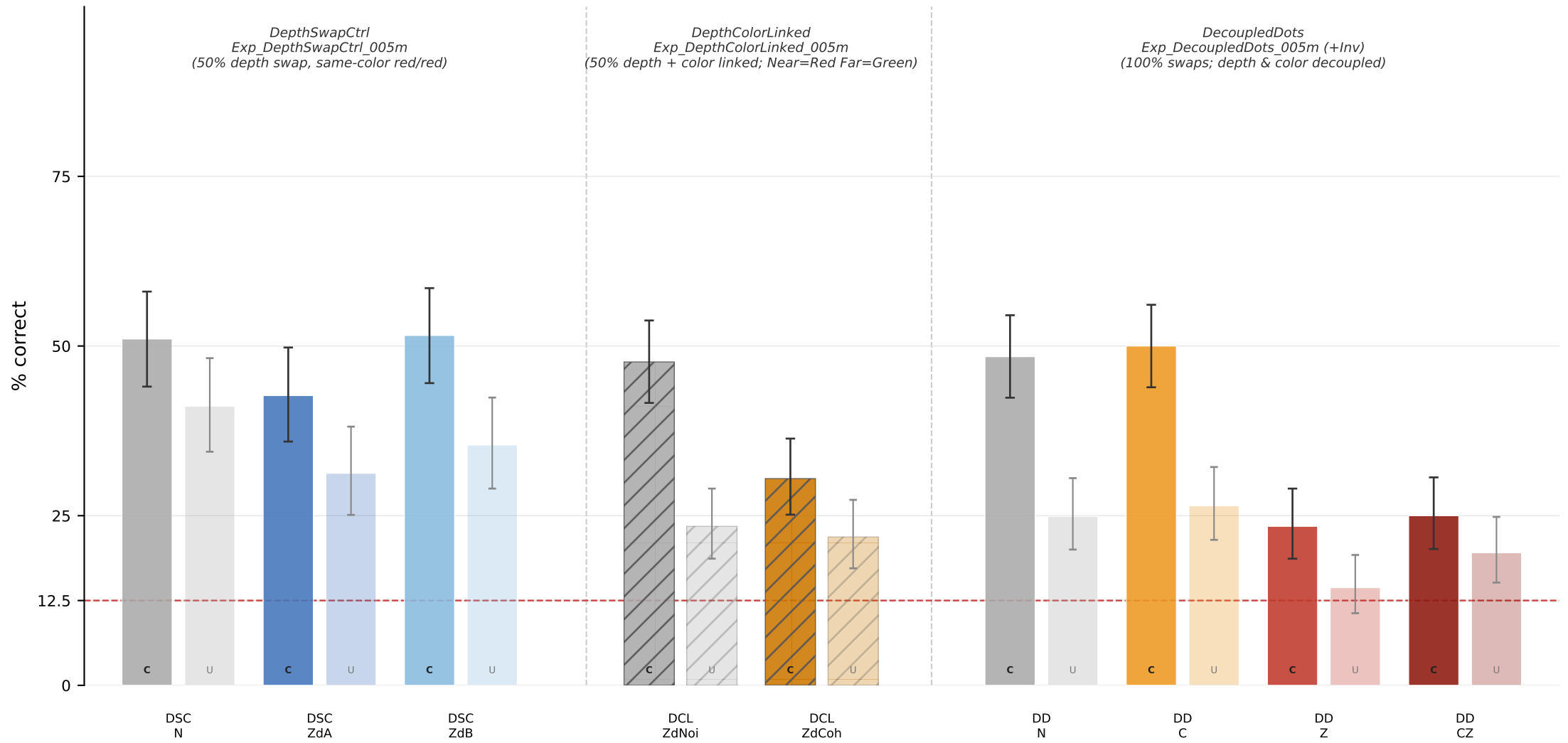


Cross-experiment depth-swap disruption · Dot-cued (solid) vs Uncued (faded) accuracy

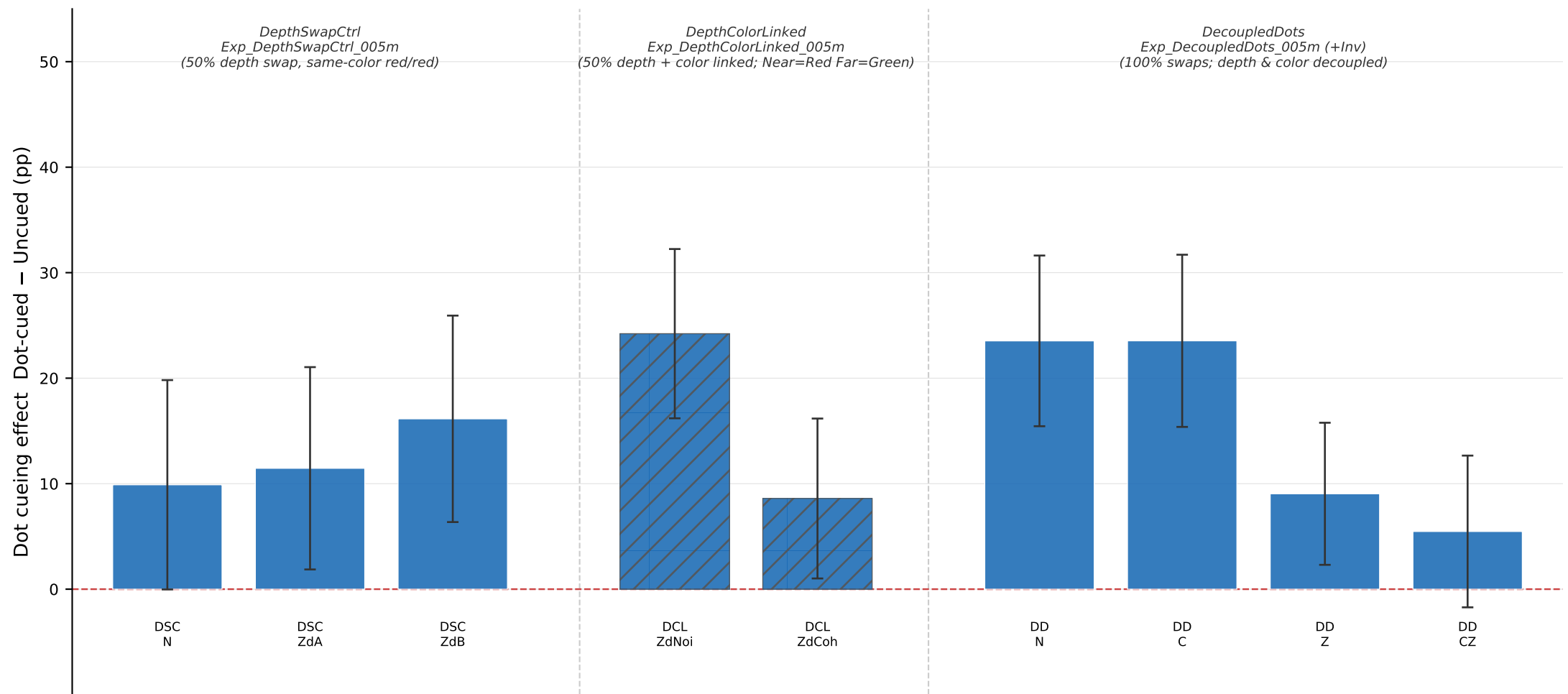


Abbreviations — Experiments: DSC = DepthSwapCtrl · DCL = DepthColorLinked · DD = DecoupledDots

Conditions: N = no swap (baseline) · C = 100% color swap only · Z = 100% depth swap only · CZ = 100% depth + color · ZdA / ZdCoh = coherent subfields swap→ translator changes depth plane · ZdB / ZdNoi = noise subfields swap→ coherent translator unchanged

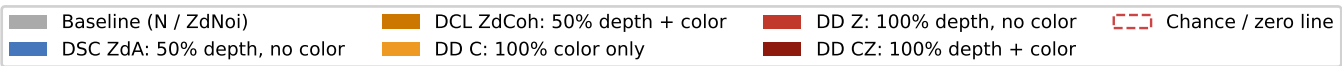


Cross-experiment depth-swap disruption · Dot cueing effect (Dot-cued – Uncued, pp)



Abbreviations — Experiments: DSC = DepthSwapCtrl · DCL = DepthColorLinked · DD = DecoupledDots

Conditions: N = no swap (baseline) · C = 100% color swap only · Z = 100% depth swap only · CZ = 100% depth + color · ZdA / ZdCoh = coherent subfields swap→ translator changes depth plane · ZdB / ZdNoi = noise subfields swap→ coherent translator unchanged



What disrupts dot cueing? — Summary of findings across three experiments

Dot cueing refers to the advantage conferred by an onset dot cue that briefly marks one of two overlapping transparent motion fields. On a given trial the "dot-cued" field later translates; the observer reports its direction. Dot cueing = (dot-cued accuracy) – (uncued accuracy).

WHAT REDUCES DOT CUEING

1. Coherent-object depth-plane change at translation onset [CRITICAL FACTOR]
When the translating (dot-cued) field changes depth plane at the moment it starts to move, dot cueing drops substantially:
- DSC ZdA (50% depth, same-color): cueing $\approx$ +12 pp n.s. (vs baseline +34 pp **)
 - DCL ZdCoh (50% depth + color, linked): cueing $\approx$ +7 pp †
 - DD Z (100% depth swap, color fixed): cueing $\approx$ +9 pp ** (vs baseline +28 pp ***)
- The translator losing its depth identity – even in just 50% of dot-cloud subfields – is sufficient to nearly abolish the cueing benefit.
2. Combined depth + color change (coherent) does not add further disruption beyond depth alone.
DD CZ $\approx$ DD Z in cueing magnitude, and DCL ZdCoh $\approx$ DSC ZdA.
Color change alone (DD C) has no effect (see below).

WHAT DOES NOT REDUCE DOT CUEING

3. Color swap alone (100%): DD C cueing $\approx$ +22 pp *** – indistinguishable from no-swap baseline.
Color identity of the translating field is irrelevant to dot cueing.
4. Noise-only depth swap (non-coherent subfields change depth):
- DSC ZdB: cueing $\approx$ +56 pp *** (enhanced relative to baseline)
 - DCL ZdNoi: cueing $\approx$ +26 pp ***
- When the coherent translator is unchanged and only non-coherent subfields swap depth, dot cueing is maintained or enhanced – the cue remains predictive.

KEY INFERENCE: DOSE DOES NOT EXPLAIN THE EFFECT

5. 50% coherent swap (DCL ZdCoh) $\approx$ 100% depth swap (DD Z) in disruption magnitude.
If disruption were proportional to the quantity of depth change in the scene, the 100% swap should be twice as disruptive as the 50% swap. It is not.
The effect tracks whether the coherent object changes depth identity, not the total amount of depth change in the stimulus.

TERMINOLOGY NOTE

Dot cueing – advantage from the onset dot marking the translating field (F1)
Depth-field cueing – advantage from the translating field occupying its pre-cue depth plane (F2)
Color-field cueing – advantage from the translating field retaining its pre-cue color (F3; null)
These three factors were dissociated in Exp_DecoupledDots_005m (GLM2: F1×F2 interaction dominates; F3 AME = +0.9 pp, p = .64).